# Week 18: Read Recognized Research, Rebuild Your Idea

## Learning from NSF CAREER Awards and Nature/Science-Family Papers

Instructor / Mentor

Kiki Research Training Program

Week 18 of the Revised 22-Week Program

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# Why Read Recognized Projects?

# Today's Big Mission

Last week, you looked inward and formed a first research idea. This week, we look outward:

**Mission**

Study how recognized researchers turn a motivation into a problem, a method, an evidence plan, a societal contribution, and a credible request for support.

Then return to your own proposal and ask:

- What became clearer?
- What should change?
- What new possibility did another project reveal?

# Learning Objectives

By the end of class, students can:

1. distinguish an award abstract, a research proposal, and a published paper;
2. extract the problem, method, data/resources, evidence, and broader impacts;
3. identify intended beneficiaries without claiming that researchers automatically "speak for" them;
4. separate text-supported statements from reasonable inferences;
5. compare recognized projects across technical and social dimensions; and
6. revise a personal research idea after learning from strong examples.

# 120-Minute Flow

1. **0–10 min:** choose which two short project pitches you would support.
2. **10–25 min:** award abstracts versus proposals versus papers.
3. **25–42 min:** learn the Six-Lens Research Map.
4. **42–47 min:** first-pass reading demonstration.
5. **47–52 min:** break.
6. **52–82 min:** three-source jigsaw reading.
7. **82–98 min:** gallery walk and cross-project comparison.
8. **98–112 min:** community audit and evidence-versus-inference check.
9. **112–120 min:** rebuild your own proposal and launch homework.

# Opening Challenge: Which Project Would You Support?

Read three 90-word project cards. You control two imaginary research grants.

1. Select two projects.
2. Write the strongest reason for each decision.
3. Name one missing fact that could reverse your decision.

**The point**

An exciting topic is not enough. Reviewers need a connected argument: important problem, credible plan, necessary resources, meaningful evidence, and responsible impact.

# Prestige Is a Starting Filter, Not a Proof Stamp

NSF CAREER awards and selective journals can expose us to ambitious, carefully reviewed work. Recognition tells us the work passed a demanding selection process.

## Learn from recognition

- how a problem is framed;
- how evidence is organized;
- how significance is argued;
- how communities and impacts are named.

## Keep thinking critically

- prestigious work can have limitations;
- peer review does not guarantee truth;
- a famous venue does not replace evidence;
- your summary may still be wrong.

# Three Documents, Three Jobs

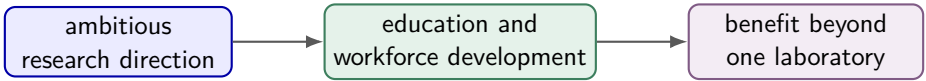
| Document | When written | Main job | What is available? |
|---|---|---|---|
| Research proposal | before the work | persuade reviewers that a plan deserves support | plan, preliminary evidence, resources, expected impacts |
| NSF award abstract | after selection, before/during work | publicly summarize an awarded plan | short description; usually not the complete confidential proposal |
| Published paper | after experiments or analysis | report and defend completed work | methods, data description, results, limitations, references |

## Language check

A proposal says what researchers *will do*. A paper reports what they *did and found*. Never turn a planned outcome into a completed result.

# What Makes NSF CAREER Special?

A CAREER project is designed around the development of an early-career faculty member as a teacher-scholar. Research and education should form one coherent plan.

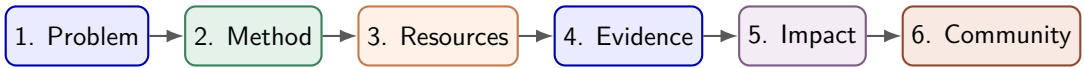



NSF: Broader Impacts guidance

# The Six-Lens Research Map

# The Six-Lens Research Map

Every group extracts the same six elements:




**One final question surrounds all six lenses**

What can this project teach or inspire me to change in my own research idea?

# Lens 1: What Problem Is Worth Solving?

Extract four connected parts:

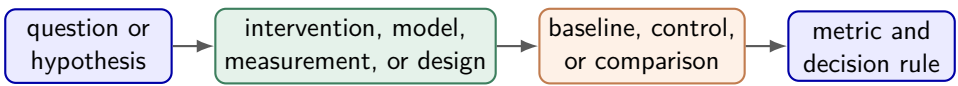
1. **Context:** where and when does the problem occur?
2. **Gap:** what is unknown, unreliable, unavailable, or unfair?
3. **Consequence:** what happens if the gap remains?
4. **Research question:** what exactly will this project determine or create?

## Compression test

Because [gap] occurs in [context], [stakeholder] experiences [consequence]. The project asks whether/how [bounded question].

# Lens 2: What Method Connects the Question to an Answer?

Do not write only "AI," "experiment," or "simulation." Identify the research logic.




A credible method explains why the resulting evidence can answer the stated question.

# Lens 3: What Data and Resources Make the Work Possible?

Search for more than a dataset name:

- observations, participants, sensors, simulations, documents, or labels;
- time range, location, population, scale, and sampling frequency;
- equipment, software, computing, laboratories, partnerships, and expertise;
- access, privacy, consent, ownership, cost, and missing-data constraints;
- the requested resource and why it is necessary.

**Feasibility question**

Could another student locate, access, and understand the required inputs from this description?

# Lens 4: What Would Count as Evidence?

**For a proposal or award abstract**

- planned experiment or analysis;
- expected comparison;
- success measure;
- promised product or capability.

**For a completed paper**

- observed result;
- effect size or model performance;
- uncertainty and robustness;
- limitation or alternative explanation.

Evidence is not the same as the authors' interpretation of the evidence.

# Lens 5: What Is the Broader Impact?

NSF describes broader impacts as the potential to benefit society and contribute to desired societal outcomes.

## Possible pathways

- safer or more reliable infrastructure;
- education and public engagement;
- broader participation in STEM;
- open data, tools, or standards;
- economic or environmental benefit.

Official NSF explanation and review criteria

## Demand a mechanism

Who does what, for whom, using which output, and how would success be observed?

# Lens 6: Which Community Benefits—and Who Has Voice?

Replace the vague question "Who does this project speak for?" with a more careful audit:

1. Who is expected to benefit directly and indirectly?
2. Who supplies data, labor, land, attention, or trust?
3. Who might face risk, cost, surveillance, or exclusion?
4. Was the community consulted, partnered with, or only described?
5. Who defines success and controls the resulting tool or data?

**Important**

A project may intend to benefit a community without representing that community's voice. Look for evidence of participation before claiming representation.

# Three Evidence Labels Prevent Overclaiming

Mark every summary statement with one label:

**TEXT** Explicitly stated in the source. Record the page, paragraph, or sentence.

**INFERENCE** Reasonable interpretation supported by clues, but not directly stated.

**QUESTION** Missing, ambiguous, or impossible to verify from the assigned excerpt.

## Example

"The award includes K–12 summer camps" is TEXT. "Local rural students helped design the camps" is a QUESTION unless the source says so.

# The One-Page Research Inspiration Canvas

| Lens | Required note |
|---|---|
| Problem | context, gap, consequence, bounded question |
| Method | action/model, comparison, metric, why it answers the question |
| Resources | data/materials, scale, access, tools, people, constraints |
| Evidence | planned evidence or completed result; uncertainty and limitations |
| Broader impact | beneficiary, output, pathway to benefit, success indicator |
| Community | benefits, burdens, participation, voice, control |
| Inspiration | one design choice to borrow; one choice to question; one change to my idea |

Every factual entry needs a location in the source.

# Three Recognized Research Cases

# Today's Reading Packet

Each group studies one source, then teaches it to classmates:

1. **NSF CAREER award:** intelligent adaptive protection for modern power systems.
2. **Science Advances paper:** whether basic off-grid electricity access changes socioeconomic outcomes.
3. **Nature Energy paper:** measuring unequal power quality and reliability in Accra, Ghana.

## Why this combination?

The cases contrast a future technical plan, a completed randomized field experiment, and a community-centered measurement framework.

# Case A: NSF CAREER—Adaptive Grid Protection

**Project:** "Stochastic Optimization and Physics-informed Machine Learning for Scalable and Intelligent Adaptive Protection of Power Systems" (NSF Award #2338555; Ali Bidram, University of New Mexico).

## Starting clues

- Modern grids change as renewable resources and loads change.
- Conventional protection settings may not adapt to prevailing conditions.
- The project proposes stochastic optimization, physics-informed ML, and hardware/software validation.

Official NSF award record  UNM project overview

## Impact clues

- grid resilience during extreme events;
- K–12 and undergraduate outreach;
- engagement of underrepresented students;
- power-engineering workforce development in New Mexico.

# Case A Team Challenge

Build a Six-Lens Map, then answer:

1. What failure of existing protection motivates the work?
2. What roles do physics, machine learning, optimization, hardware, and software each play?
3. Which resources would be difficult for a high-school project to reproduce?
4. What evidence would convince you that protection is truly more adaptive and resilient?
5. Are technical reliability and education separate impacts, or can they reinforce each other?
6. Which communities are named? What evidence of co-design is present or absent?

**Tense check**

This is an awarded plan. Write “the project proposes”—not “the researchers proved.”

# Case B: Science Advances—Basic Electricity Access

**Paper:** Aklin et al., "Does basic energy access generate socioeconomic benefits? A field experiment with off-grid solar power in India," *Science Advances* (2017).

## Design clues

- randomized field experiment;
- 1,281 rural households;
- basic off-grid solar electricity intervention;
- comparison of treated and control households.

## Interpretation challenge

The study separates improvements in electricity access from larger claims about income, education, health, or other socioeconomic outcomes.

## Ask

What can the experiment establish, and what remains outside its intervention, duration, or measures?

Official article and abstract

# Case B Team Challenge

Build a Six-Lens Map, then answer:

1. What is the difference between electricity *access* and socioeconomic development?
2. Why is random assignment valuable for this question?
3. What did households, field teams, funders, and technology providers contribute?
4. Which outcomes were measured, and over what time horizon?
5. What conclusion is supported? What stronger conclusion would be an overclaim?
6. Who benefits, who participates as a research subject, and who decides which outcomes matter?

## Transferable lesson

A useful intervention can succeed at its immediate objective without automatically transforming every long-term social outcome.

# Case C: Nature Energy—Power Quality Inequities

**Paper:** Osunmuyiwa et al., "Mobilizing power quality and reliability measurements for electricity equity and justice in Africa," *Nature Energy* (2025).

## Problem and resource clues

- connection statistics can hide poor service quality;
- granular power-quality data are scarce;
- the demonstration uses one year of remotely sensed data from Accra, Ghana.

Official article and abstract

## Method and community clues

- links power-system measures with energy justice, capability, and poverty concepts;
- studies neighborhoods, household capabilities, and multidimensional poverty;
- aims to guide more equitable infrastructure decisions.

# Case C Team Challenge

Build a Six-Lens Map, then answer:

1. Why can "connected to the grid" be a misleading success measure?
2. How does the study combine technical measurements with social concepts?
3. What must be measured remotely, and what requires household or neighborhood knowledge?
4. Which observed relationships are not automatically causal?
5. How might the framework change where infrastructure investment goes?
6. What would genuine community voice look like beyond measuring a community?

## Transferable lesson

Choosing a measurement is also choosing which experiences become visible to decision makers.

# Optional Technical Companion: An Open Grid Dataset

**Paper:** Zheng et al., "A multi-scale time-series dataset with benchmark for machine learning in decarbonized energy grids," *Scientific Data* (2022).

- Builds an open dataset spanning load, weather, renewable generation, voltage, current, and simulated disturbances.
- Combines real-world time series with transmission–distribution co-simulation.
- Defines benchmark tasks for disturbance detection, forecasting, and synthetic time-series generation.
- Opens a technical field to ML researchers who lack access to confidential operational grid data.

## Connection to the Six-Lens Map

Sometimes the main research contribution is a shared resource that allows an entire community to ask better questions.

Open-access Nature Portfolio article

# Wrap-Up and Instructor Notes

# Exit Ticket

Answer in one sentence each:

1. What is one difference between an NSF award abstract and a paper?
2. Why must a method include a comparison or evaluation rule?
3. What resource was essential in one case today?
4. What turns a hoped-for benefit into a credible broader-impact pathway?
5. Why is "the project speaks for this community" often too strong?
6. Which recognized project changed your thinking most, and what will you revise?

# Instructor Preparation Checklist

Before class:

1. print or share the official abstract/project-summary pages;
2. number paragraphs or sentences for fast evidence citation;
3. prepare one completed Six-Lens example and three intentionally weak summaries;
4. assign reading groups based on difficulty and student interest;
5. prepare poster paper or a shared six-column workspace;
6. collect each student's Week 17 proposal seed for the final revision activity;
7. provide accessible alternatives for paywalled or technically dense papers.

# Core Reading Links

- NSF Award #2338555: adaptive protection of power systems
- NSF: Broader Impacts overview and review questions
- Science Advances: off-grid solar field experiment in India
- Nature Energy: power quality, reliability, and electricity equity
- Scientific Data: open multi-scale grid dataset and ML benchmarks

## Reading order for a 9th-grade student

Title → abstract → figures and captions → introduction ending → methods overview → discussion and limitations.

# Key Takeaways

1. Recognized research is useful for learning structure, not for suspending judgment.
2. Read proposals as plans and papers as reports of completed evidence.
3. Use six lenses: problem, method, resources, evidence, broader impact, and community.
4. Separate what the source states from what you infer or still need to ask.
5. Intended benefit does not automatically equal community voice or representation.
6. Inspiration means adapting a strong research choice to your own motivation and constraints.
7. A better second idea is often not completely new—it is more connected, testable, feasible, and responsible.

# Read deeply. Question carefully. Rebuild boldly.

Next checkpoint: Proposal Seed v0.2 and the evidence/data plan.